

Objective

The objective of this task is to develop a **Prime Number Analyzer** using Python and Streamlit. The application checks whether a given number is prime and generates all prime numbers within a user-specified range. The task helps strengthen understanding of loops, conditional statements, functions, mathematical logic, and basic algorithm optimization.

Key Features

- Checks whether a number is prime.
- Generates prime numbers within a specified range.
- Uses a reusable `is_prime()` function.
- Checks divisibility only up to the square root of the number.
- Handles 0 and 1 correctly as non-prime numbers.
- Handles the special case of 2.
- Displays all generated prime numbers.
- Validates the starting and ending range.
- Includes an explanation of the prime-checking algorithm.
- Provides an interactive Streamlit interface.

Technologies Used

The project uses **Python** as the programming language and **Streamlit** for the graphical web interface. Python functions, loops, conditional statements, the `math.isqrt()` function, and the modulo operator `%` are used to implement the prime-number checking logic.

Learning Outcomes

This task helps develop practical knowledge of Python functions, loops, conditional logic, mathematical operations, and algorithmic thinking. It also introduces a more efficient prime-checking approach by limiting divisibility checks to values up to the square root of the number.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY13>

```

    '</div>',
    unsafe_allow_html=True
)

st.divider()

def is_prime(number):
    """
    Returns True if the number is prime,
    otherwise returns False.
    """

    # Numbers less than 2 are not prime
    if number < 2:
        return False

    # 2 is the only even prime number
    if number == 2:
        return True

    # Other even numbers are not prime
    if number % 2 == 0:
        return False

    # Check only odd divisors up to square root
    limit = math.isqrt(number)
    for divisor in range(3, limit + 1, 2):
        if number % divisor == 0:
            return False

    return True

st.markdown(
    '<div class="section-title">🔍 Prime Number Checker</div>',
    unsafe_allow_html=True
)

number = st.number_input(
    "Enter a number:",
    min_value=0,
    step=1,
    value=17
)

if st.button("Check Prime", use_container_width=True):
    number = int(number)

```

```

app.py
1 import streamlit as st
2 import math
3
4 st.set_page_config(
5     page_title="Prime Number Analyzer",
6     page_icon="🔢",
7     layout="centered"
8 )
9
10 st.markdown("""
11 <style>
12     .main-title {
13         text-align: center;
14         font-size: 40px;
15         font-weight: bold;
16     }
17
18     .subtitle {
19         text-align: center;
20         font-size: 18px;
21         margin-bottom: 25px;
22     }
23
24     .prime-result {
25         padding: 18px;
26         border-radius: 10px;
27         text-align: center;
28         font-size: 22px;
29         font-weight: bold;
30     }
31
32     .section-title {
33         font-size: 24px;
34         font-weight: bold;
35     }
36 </style>
37 """, unsafe_allow_html=True)
38
39
40 st.markdown(
41     '<div class="main-title">🔢 Prime Number Analyzer</div>',
42     unsafe_allow_html=True
43 )
44
45 st.markdown(
46     '<div class="subtitle">'
47     'Check prime numbers and generate primes within a range'

```

OUTPUT: <http://localhost:8501>

1234

Prime Number Analyzer

Check prime numbers and generate primes within a range

🔍 Prime Number Checker

Enter a number:

61

Check Prime

🎉 61 is a PRIME number.

✅ 61 is PRIME

🔢 Prime Number Range Generator

Starting number:

1

Ending number:

50

Generate Prime Numbers

🔢 Prime Number Range Generator

Starting number:

1

Ending number:

50

Generate Prime Numbers

💡 Examples

✅ 2 → Prime

✅ 7 → Prime

✅ 13 → Prime

❌ 1 → Not Prime

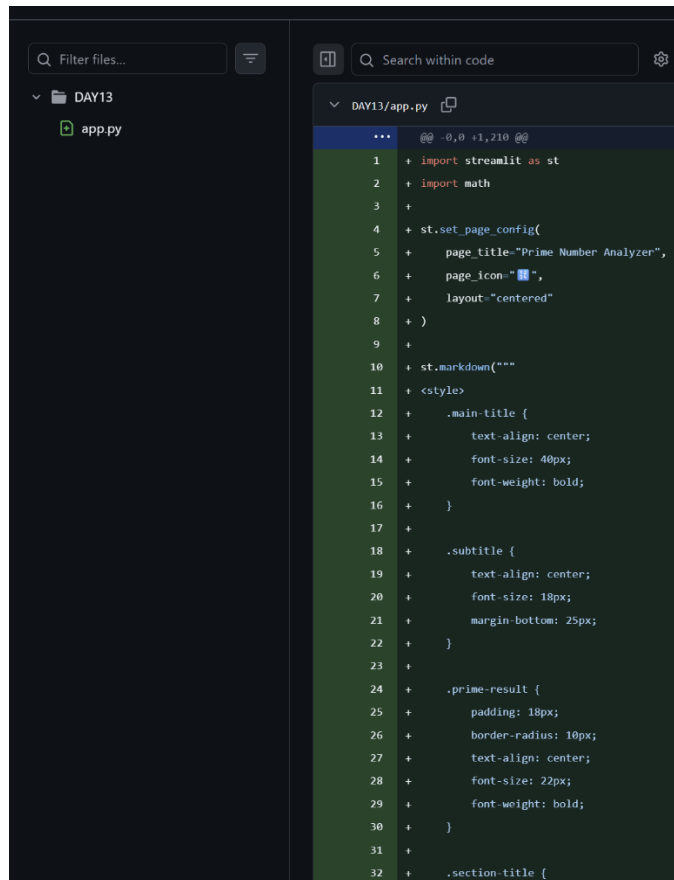
❌ 10 → Not Prime

❌ 0 → Not Prime

🧠 Algorithm

A number is prime if it is greater than 1 and has no divisors other than 1 and itself. The program first handles values below 2 and the special case of 2. It then eliminates other even numbers and checks only odd divisors up to the square root of the number. This avoids unnecessary divisibility checks and makes the prime-checking process more efficient.

Github: <https://github.com/shambhavi0804/VEDA-internship/commit/b899bce6642e94513ef47c6d>



```
... @@ -0,0 +1,210 @@
1 + import streamlit as st
2 + import math
3 +
4 + st.set_page_config(
5 +     page_title="Prime Number Analyzer",
6 +     page_icon="🔢",
7 +     layout="centered"
8 + )
9 +
10 + st.markdown("""
11 + <style>
12 +     .main-title {
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27 +         text-align: center;
28 +         font-size: 22px;
29 +         font-weight: bold;
30 +     }
31 +
32 +     .section-title {
```

Linkedin: <https://lnkd.in/p/dsWycU4U>

